

STATEMENT OF WORK

High Output Neutron Generator

4 February 2025

1.0. **BACKGROUND:** A neutron generator uses the fusion of either two deuterium atoms or a deuterium and tritium atom to output neutrons at high energy. High-output generators are capable of providing up to 1×10^{10} neutrons per second. A source of neutrons with this output allows for a greater capacity for measurements during research and academic courses due to lower required measurement time. Additionally, the neutrons that are generated are high energy neutrons and allow for a wide range of study from spectrum identification to radiation hardening experiments. The capabilities provided by this generator is of great interest to other organizations who could provide funding for research projects. The direct benefit of this generator is to students in the following NENG courses:
NENG612 - Nuclear Engineering Laboratory
NENG650 - Nuclear Instrumentation
NENG812 - Advanced Nuclear Engineering Laboratory

2.0. **SCOPE:** The purchase of high output neutron generator

3.0. **REQUIREMENTS:** The contractor shall provide the following:

- 3.1. Neutron generator with the following constraints
 - 3.1.1. Performance standards
 - 3.1.1.1. Consistent, maximum neutron output of $1 \cdot 10^{10} \text{ n/s}$
 - 3.1.1.2. Output stability of 5% after short (<2 hr) warm up time
 - 3.1.1.3. Live time of >95%
 - 3.1.1.4. Operates at room temperature
 - 3.1.2. Dimensions
 - 3.1.2.1. Less than 1.5 meters in length
 - 3.1.2.2. Less than 0.5 meters in width
 - 3.1.2.3. Less than 0.5 meters in height
 - 3.1.2.4. (Generator should fit on a heavy-duty tabletop)
 - 3.1.3. Weight
 - 3.1.3.1. Less than 100kg
- 3.2. Ancillary equipment;
 - 3.2.1. Control electronics, which should be contained in a single rack that can be in a different location than the generator itself.
 - 3.2.2. Target heat exchanger
 - 3.2.3. Roughing Pump
 - 3.2.4. Control computer with software to run the equipment and store data
 - 3.2.5. Manuals for all equipment
- 3.3. On site (Wright-Patterson AFB) installation assistance and training, 2 people, up to 5-day trip
- 3.4. Maintenance Contract for 1 year (starting the year after purchase), including

- 3.4.1. Replacing any broken parts
- 3.4.2. Confirming performance of the system
- 3.4.3. Installing any potential hardware upgrades
- 3.4.4. Installing any potential software upgrades
- 3.5. Ground shipping to Wright-Patterson AFB

4.0. Period of Performance:

- 4.1. The vendor shall deliver and complete all requirements of this Statement of Work no later than 8 months after contract award
- 4.2. Contractor shall be responsible for transportation of all new products, loading, unloading, storage, and all necessary movement from factory to final onsite set up.
- 4.3. The contractor Project Manager shall be responsible for on-site coordination of all deliveries of their product, to include access to and operation of any equipment needed for unloading and movement of materials or product necessary to complete all requirements of this Statement of Work.
- 4.4. Free from defects in material and workmanship, under normal and proper use in accordance with equipment user manual, for the period of time equal to 1 year after receipt (this encompasses the initial one year hardware warranty).

5.0. Place of Performance:

Air Force Institute of Technology (AFIT/ENP)
ATTN: Mr. Greg Smith
2950 Hobson Way, Building 640, Room 265,
Wright-Patterson AFB, OH 45433-7765

- 6.0. **Payment:** 100% payment upon completion of all requirements specified in this Statement of Work. Payment shall be made on a net 30 basis. Invoices shall be submitted and paid through Wide Area Workflow.